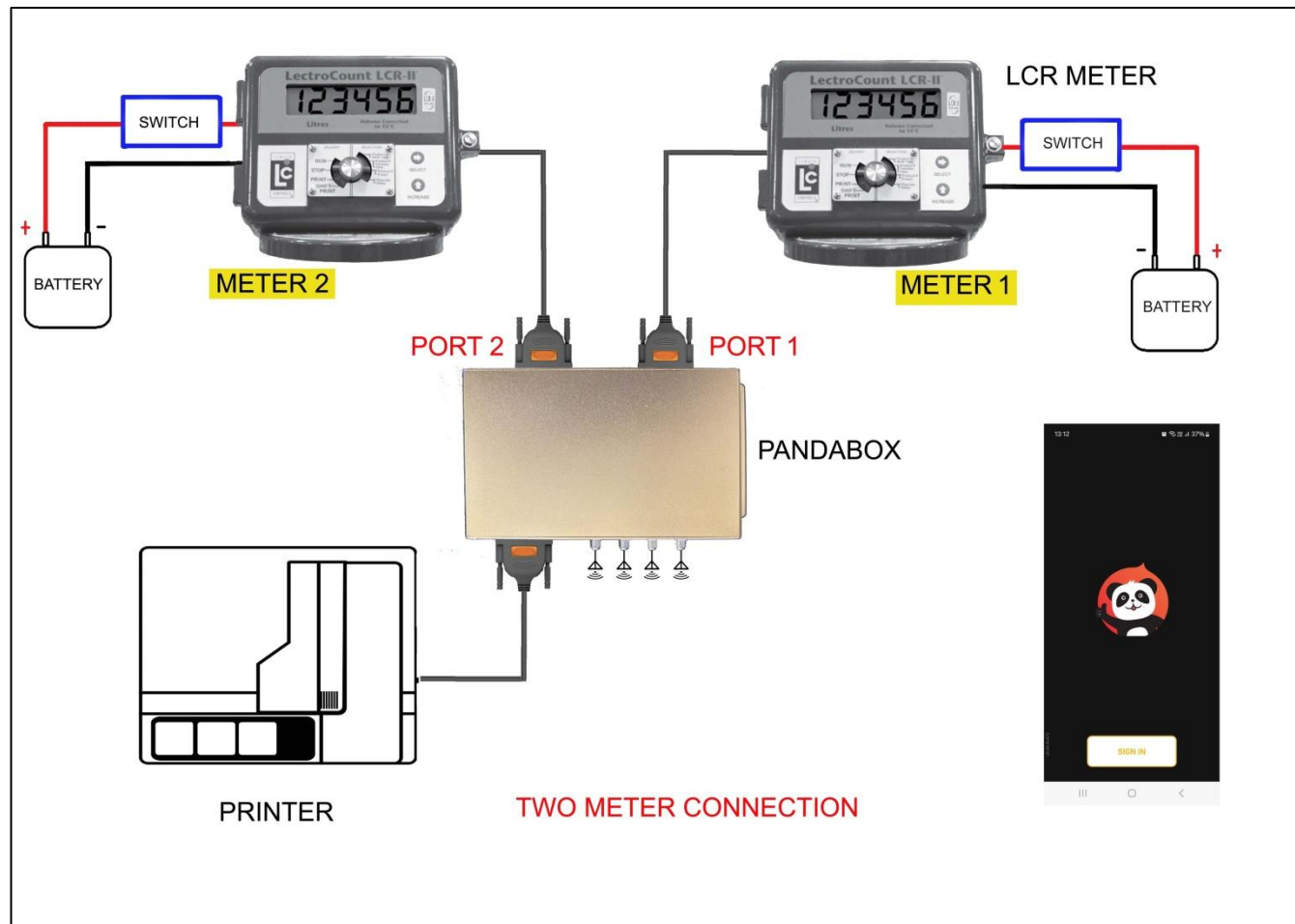


PandaBox

How it works.

PandaBox SetUp



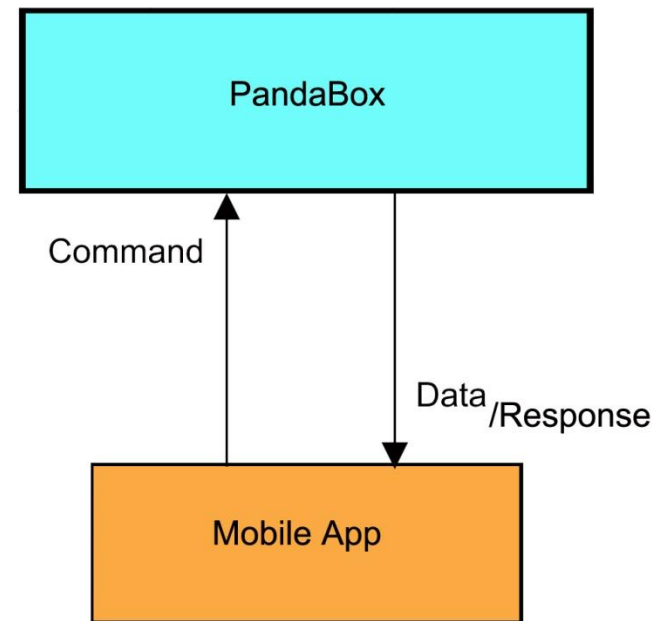
Data Flow Using PandaBox

1. Pairing:

- i. The user initiates the connection between the Mobile App and the PandaBox by pairing the handheld device with the Box via Bluetooth connection.

2. Box Connection:

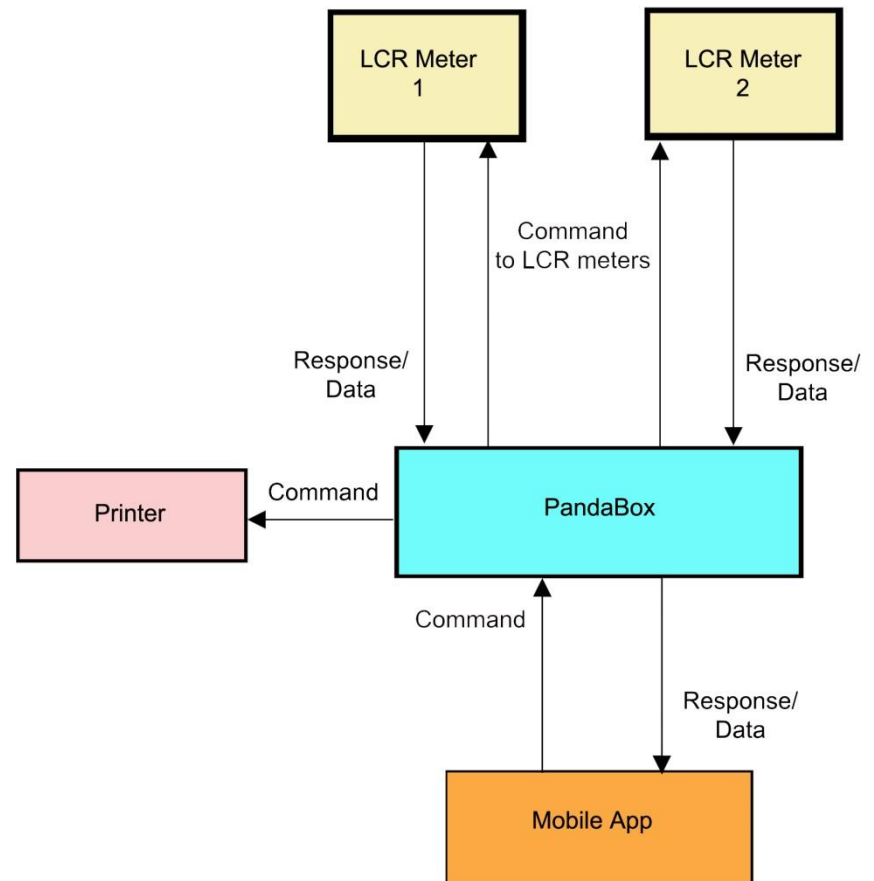
- i. When the Mobile App connects to the Box, it sets LCP mode to the LCR meters. We call this the Data mode. Due to this you will not be able to use the LCR meters in Manual mode.
- ii. As the Data Mode is initiated, the mobile app synchronizes the Handheld device's time with the PandaBox. It follows Epoch timestamp.



Data Flow Using PandaBox

2. Box Connection:

- iii. The app then sets the port LCP node address to the two available ports in the Box.
This should match the LCP node address for individual meters which are updated to the register information in the Fleetpanda Web App.
If the node addresses are valid, the App will then send a command to reset the meters and retrieve the totalizer value of individual meter via the Box.
The totalizer value for either meter and the displayed value on the individual meters are then stored in the PandaBox. The Box can store the values for a weeks time. Then after



Data Flow Using PandaBox

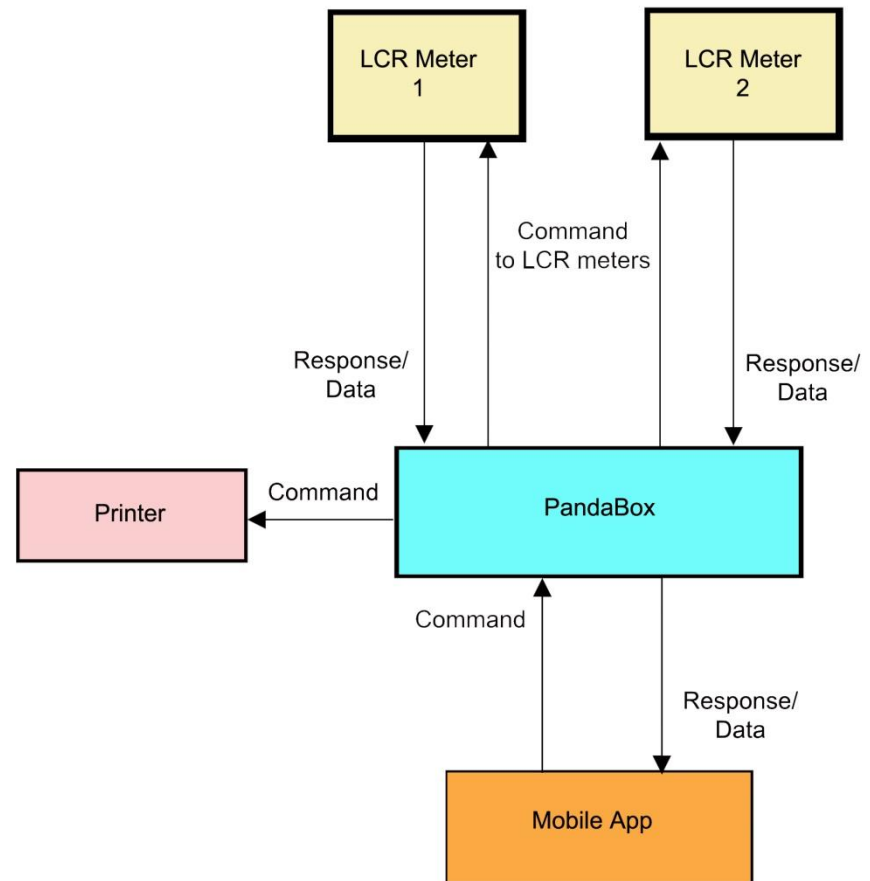
3. Data Retrieval

The PandaBox stores the delivered gallons' values for individual LCR meters and is fed to the Mobile App upon request. The data can be read by the Mobile App in two different ways.

i. Within Range (Online mode)

- In this environment, the mobile app is consistently connected to the PandaBox via Bluetooth. The data is reflected in the display of the handheld device in the real time. It will though take atleast 0.6 seconds to reflect the actual value seen on the display of the LCR meter to the display screen of the handheld device.

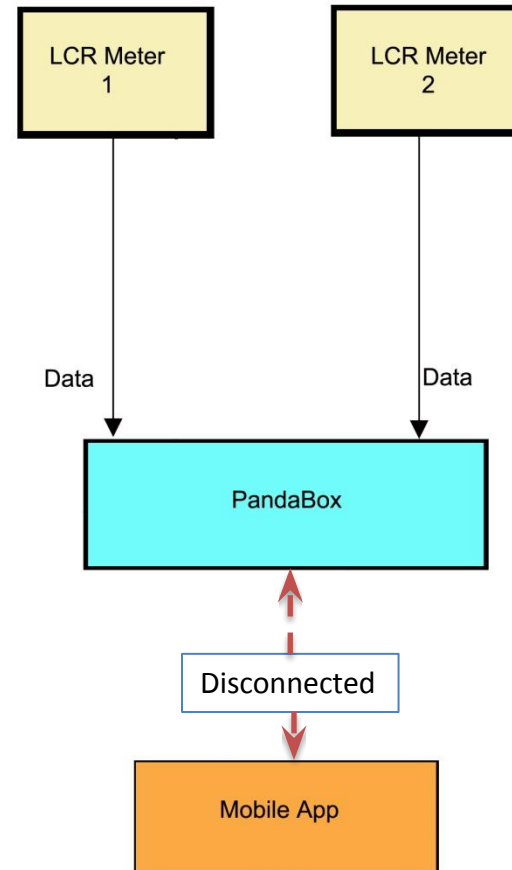
In this state, the mobile app can start, pause and stop the LCR meters and hence when a printer is connected to the pandabox, ticket can be issued at will of the user.



Data Flow Using PandaBox

3. Data Retrieval

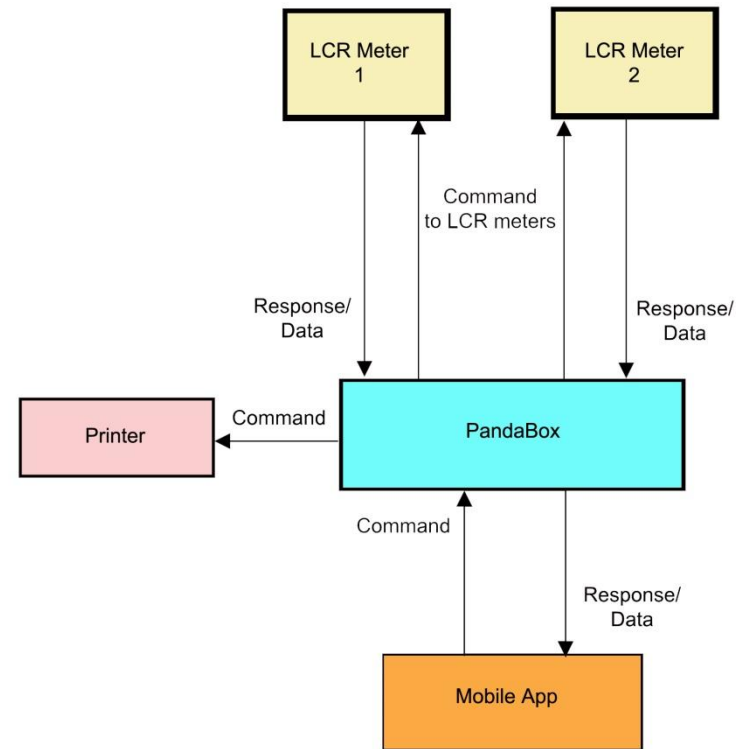
- ii. Outside Range (Offline mode)
 - In this environment, the bluetooth connection between the handheld device and the PandaBox is broken. Hence real time data acquisition through the mobile app cannot be achieved.
 - The LCR meters though are active in this state as well, hence, delivery of fuel can be made through the meters.
 - The delivered gallon values, in this state, are stored in the PandaBox itself.
 - One can deliver to any number of assets during this state, and all of the values will be stored in the memory device in Pandabox.
 - The LCR meters are in the Run state only and hence ticket printing is not available.



Data Flow Using PandaBox

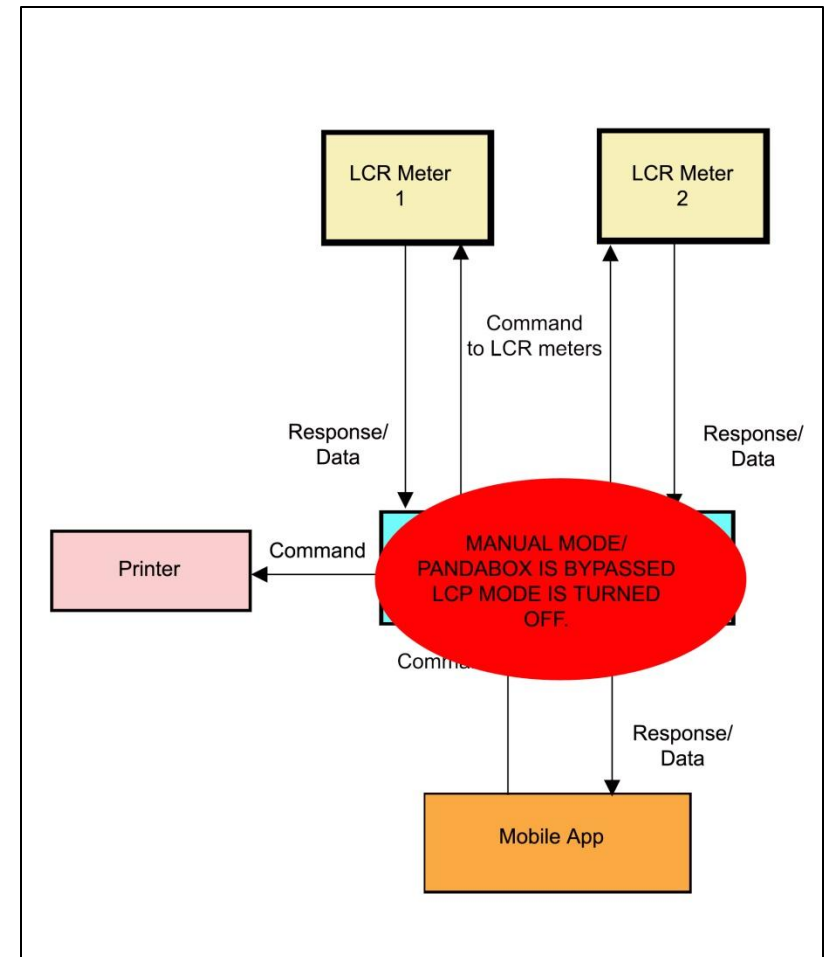
3. Data Retrieval

- ii. Outside Range (Offline mode)
 - Once the user feels like he or she has delivered to all the asset for any particular customer, they must establish the bluetooth connection between handheld device and the PandaBox. So basically, the user should move closer to the box, which will initiate the Box reconnection.
- When the bluetooth connection is re-established, mobile app requests the PandaBox to provide the values for the independent assets. This is possible by computing the different timestamps when those individual assets were selected and the data stored during those timestamps in the PandaBox.



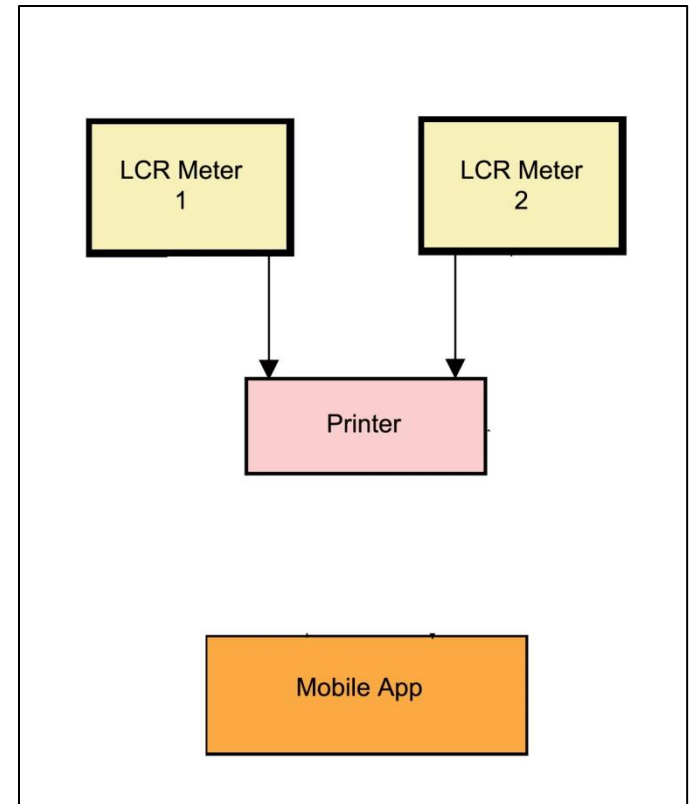
Manual Mode

1. If there is a need to shift to manual mode during and ongoing delivery, it is imperative that we turn OFF the LCP mode assigned to the LCR meters.
2. To do that, we need to go into the settings in the Mobile App and switch to Manual mode.
3. With the Manual mode 'ON', please power cycle the LCR meters.
This is due to the feature imposed by the Liquid control such that the LCR meters must be power cycled when the LCP modes are changed to manual.
4. The users can now use the LCR meters in Manual mode and are not under the influence of the Mobile App or PandaBox.



Manual Mode

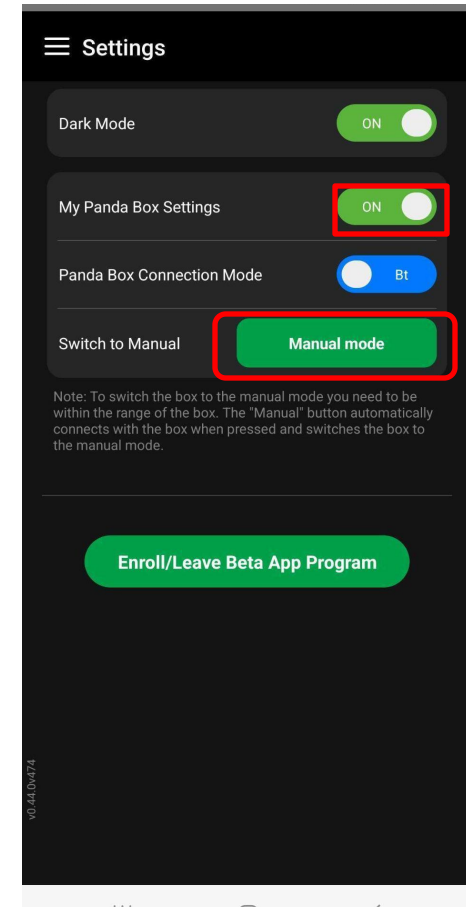
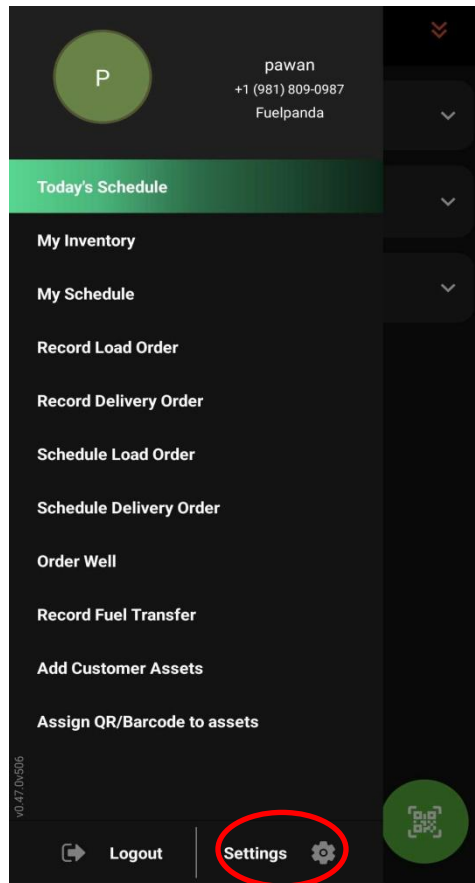
5. Ensure the Printers are connected via the lappad adapter if ticket printing is required.
6. If you feel like using the PandaBox, you can simply turn on the Pandabox settings to ON position.
7. If you intend not to use PandaBox or circumstances occur when the PandaBox is not responsive (which will seldom occur), then rather than going to 'Settings' to switch to manual mode, you can unplug the data cables connected to the PandaBox and power cycle the LCR meters.
This will again allow you to use the LCR meters manually.



Switching **Manual Mode** Using the PandaBox

Steps:

1. Tap on the Settings of the Mobile App.
2. In the Settings, press on 'Manual mode' that allows you to switch to manual mode.



Manual Mode Confirmation

- When the manual mode is attained, you will see the dialog boxes as below and 'My panda Box Settings' is automatically disabled.

